

THE COMPLETE FIBRE TABLE AT $n = 41$ FOR THE TWO-TO-ONE TUNNELL MAP

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THE REPRESENTATION SETS

This note accompanies the manuscript *Stable Matching and a Two-to-One Map for Tunnell's Ternary Forms*. We spell out the complete map at $n = 41$ in the original ternary coordinates. Thus

$$\mathcal{B}_{41} = \{(x, y, z) \in \mathbb{Z}^3 : 2x^2 + y^2 + 8z^2 = 41\}, \quad \mathcal{A}_{41} = \{(u, v, w) \in \mathbb{Z}^3 : 2u^2 + v^2 + 32w^2 = 41\}.$$

There are 32 points in $\mathcal{B}_{41}$ and 16 points in $\mathcal{A}_{41}$. The tables below list every target and its two preimages, so all points of both representation sets occur.

THE THREE RULES USED IN THE TABLE

Even branch. For every $(u, v, w) \in \mathcal{A}_{41}$, the first preimage is

$$(u, v, 2w) \mapsto (u, v, w).$$

This is equation (4) of the paper and is the upper bijection in the proof of Theorem 1.2.

Direct odd branch. All eight direct odd points at $n = 41$ lie in the third quarter-turn domain. Equation (10) gives

$$\Lambda_3(x, y, z) = \left(2z, y, -\frac{x}{4}\right).$$

The domain test is Proposition 3.1, and the fact that this is a bijection onto the direct target set is Theorem 3.2.

Residual odd branch. The remaining eight odd points are paired by the unique stable matching of Proposition 4.5. Equation (12) fixes the sign when the orbit matching is lifted back to individual points. Algorithm 7.1 computes the same matching, its correctness is Theorem 7.3, and the complete $n = 41$ residual run is Example 8.1. The four canonical residual pairs in original coordinates are

$$\begin{aligned} (-2, -5, -1) &\mapsto (4, 3, 0), & (-2, -5, 1) &\mapsto (4, -3, 0), \\ (-2, 5, -1) &\mapsto (0, -3, 1), & (-2, 5, 1) &\mapsto (0, -3, -1). \end{aligned}$$

Their negatives give the other four residual pairs.

DIRECT FIBRES

In each row, the middle point is the even preimage supplied by equation (4), and the last point is the odd preimage supplied by Theorem 3.2 and equation (10).

target $Y \in \mathcal{A}_{41}$	even preimage	odd preimage	direct computation
$(-2, -1, -1)$	$(-2, -1, -2)$	$(4, -1, -1)$	$\Lambda_3(4, -1, -1) = Y$
$(-2, -1, 1)$	$(-2, -1, 2)$	$(-4, -1, -1)$	$\Lambda_3(-4, -1, -1) = Y$
$(-2, 1, -1)$	$(-2, 1, -2)$	$(4, 1, -1)$	$\Lambda_3(4, 1, -1) = Y$
$(-2, 1, 1)$	$(-2, 1, 2)$	$(-4, 1, -1)$	$\Lambda_3(-4, 1, -1) = Y$
$(2, -1, -1)$	$(2, -1, -2)$	$(4, -1, 1)$	$\Lambda_3(4, -1, 1) = Y$
$(2, -1, 1)$	$(2, -1, 2)$	$(-4, -1, 1)$	$\Lambda_3(-4, -1, 1) = Y$
$(2, 1, -1)$	$(2, 1, -2)$	$(4, 1, 1)$	$\Lambda_3(4, 1, 1) = Y$
$(2, 1, 1)$	$(2, 1, 2)$	$(-4, 1, 1)$	$\Lambda_3(-4, 1, 1) = Y$

For example,

$$\Phi_{41}^{-1}(2, -1, 1) = \{(2, -1, 2), (-4, -1, 1)\}.$$

The first point maps by equation (4), while $\Lambda_3(-4, -1, 1) = (2, -1, 1)$ by equation (10). This is the direct fibre worked out in the paper immediately after the proof of Theorem 1.2.

RESIDUAL FIBRES

The notation s_i refers to the four canonical source orbits in Example 8.1. A minus sign indicates that equation (12) applies the antipodal lift to the corresponding canonical pair.

target $Y \in \mathcal{A}_{41}$	even preimage	odd preimage	residual origin
$(-4, -3, 0)$	$(-4, -3, 0)$	$(2, 5, 1)$	negative of the s_1 pair
$(-4, 3, 0)$	$(-4, 3, 0)$	$(2, 5, -1)$	negative of the s_2 pair
$(0, -3, -1)$	$(0, -3, -2)$	$(-2, 5, 1)$	s_4 pair
$(0, -3, 1)$	$(0, -3, 2)$	$(-2, 5, -1)$	s_3 pair
$(0, 3, -1)$	$(0, 3, -2)$	$(2, -5, 1)$	negative of the s_3 pair
$(0, 3, 1)$	$(0, 3, 2)$	$(2, -5, -1)$	negative of the s_4 pair
$(4, -3, 0)$	$(4, -3, 0)$	$(-2, -5, 1)$	s_2 pair
$(4, 3, 0)$	$(4, 3, 0)$	$(-2, -5, -1)$	s_1 pair

The stable matching in Example 8.1 is obtained from the proposal sequence

$$\begin{aligned} s_1 \rightarrow t_1, \quad s_2 \rightarrow t_1 \text{ (rejected)}, \quad s_3 \rightarrow t_4, \\ s_4 \rightarrow t_3, \quad s_2 \rightarrow t_4 \text{ (rejected)}, \quad s_2 \rightarrow t_2. \end{aligned}$$

It therefore matches the four source orbits as

$$s_1 \leftrightarrow t_1, \quad s_2 \leftrightarrow t_2, \quad s_3 \leftrightarrow t_4, \quad s_4 \leftrightarrow t_3,$$

after which equation (12) determines the individual signs displayed in the table.

EXHAUSTIVENESS CHECK

The direct table contains the eight targets in the image of the quarter-turn layer. The residual table contains the other eight targets. Their even preimages are the 16 points of $\mathcal{B}_{41}$ having even third coordinate; their odd preimages consist of the eight direct points and the eight residual points. Thus the tables contain 16 distinct targets and 32 distinct sources, and each target has exactly two preimages, as asserted by Theorem 1.2.

The table was also regenerated from the formalized executable map `tunnellMapTotal41`. The Lean declarations `card_BRep_41`, `card_ARep_41`, and `tunnellMapTotal41_exactly_two` verify respectively that the source and target cardinalities are 32 and 16 and that every displayed target has exactly two preimages.

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